Additional file 1: Supplementary methods and results

Split-Dependent Transfer in Multi-Target Bioactivity Prediction: A Leakage-Controlled ChEMBL 37 Benchmark and Locked Plant-Bag Extension

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1 Reproducibility scope

This supplement reports only experiments derived from the public, version-pinned ChEMBL 37 SQLite release. The VietHerb-dependent multiple-instance learning (MIL) arms are implementation-tested but prospectively locked; no synthetic or restricted row contributes to a scientific estimate. All five-seed experiments used seeds 13, 37, 71, 101, and 137. Each run manifest records dataset, metric, prediction, and configuration hashes. The machine-readable tables distributed beside this PDF contain the unrounded values underlying the manuscript. The software and evidence archive for version 0.2.0 is available at `https://doi.org/10.5281/zenodo.23537107`.

2 Cohort construction

The source archive was `chembl_37_sqlite.tar.gz`, DOI 10.6019/CHEMBL.database.37. Its SHA-256 begins `33c203740555f960`; the extracted SQLite digest begins `4be13df3b68e25dc`. Full digests are stored in the machine-readable extraction and evidence manifests. Target identity was asserted against single human proteins and UniProt entries P22303 (AChE), P35354 (PTGS2), and P27487 (DPP4).

Table 1: ChEMBL 37 extraction flow before structure aggregation.

Filter	Before	After	Excluded
Activity type IC_{50}/K_i	43,843	27,980	15,863
Binding assay	27,980	27,634	346
Units nM	27,634	23,265	4,369
Positive value	23,265	23,210	55
Valid pChEMBL	23,210	19,232	3,978
Not a potential duplicate	19,232	16,947	2,285
Resolved structure	16,947	16,941	6

All retained relations were equality measurements. Exact standardized structure–target replicates were summarized by median pChEMBL. A pair whose replicates spanned at least one log unit was contradictory. The primary binary cohort excluded 435 contradictory and 3,353 gray-zone pairs; activity was pChEMBL ≥ 6 and inactivity was pChEMBL ≤ 5. The continuous cohort excluded the same 435 contradictory pairs but retained the gray zone.

Table 2: Dataset and split audit summary.

Cohort	Molecules	Pairs	Split counts	Max similarity
Chemical-OOD binary	10,686	10,698	7,480/1,603/1,603	0.698
Chemical-OOD continuous	14,029	14,051	9,820/2,105/2,104	0.699
Temporal binary	10,697	10,708	7,896/1,734/1,067	1.000

The primary and continuous partitions used connected components at Morgan Tanimoto ≥ 0.70 and had zero standardized-structure, InChIKey, Murcko scaffold, and component overlap. The temporal cohort used the first source year: training through 2017, validation from 2018 through 2021, and test from 2022 through 2025. It excluded 60 rows without year information and censored 343 future observations from training or validation aggregation. Chemical overlap was allowed by design in this chronological sensitivity.

3 Models and selection

All fingerprint models used 2,048-bit chiral radius-2 Morgan fingerprints. Neural MLPs used hidden widths 1,024 and 512, dropout 0.25, AdamW at learning rate 3×10^{-4} and weight decay 10^{-4}, batch size 256, and patience 12. Logistic regression was L2-regularized. Extra Trees and logistic models were fit per target. Chemprop 2.2.1 consumed the same split column and masked missing labels, with a maximum of 100 epochs and patience 12.

Every tunable candidate was ranked by the mean validation objective over the prespecified seeds. The test partition was evaluated only after the selected configuration or best validation checkpoint was restored. Chemprop used validation loss for deterministic checkpoint selection because exact-seed CUDA execution made its online PRC metric incompatible with deterministic algorithms; the sealed-test PR-AUC and all secondary metrics were computed independently from stored predictions. Chemprop class-balanced sampling was disabled because the sampler produced zero batches for sparse multitask rows.

4 Primary chemical-OOD results

Table 3: Per-target PR-AUC, mean ± standard deviation over five seeds.

Model	AChE	DPP4	PTGS2
Extra Trees	0.8946 ± 0.0019	0.9943 ± 0.0006	0.9385 ± 0.0019
Logistic regression	0.8858 ± 0.0000	0.9936 ± 0.0000	0.9208 ± 0.0000
Separate MLP	0.8781 ± 0.0096	0.9844 ± 0.0035	0.9217 ± 0.0030
Shared MLP	0.8509 ± 0.0104	0.9889 ± 0.0013	0.9212 ± 0.0042
Chemprop D-MPNN	0.8445 ± 0.0126	0.9572 ± 0.0019	0.8796 ± 0.0082

The interval procedure used 1,000 hierarchical bootstrap replicates over model seed and chemical component. An empirical two-sided value of zero means no bootstrap replicate crossed zero; it is reported as $p < 0.001$, not as an exact probability of zero. Reliability-bin counts and confidence/accuracy sums are in `table_reliability_bins.csv`.

Table 4: Paired shared-minus-separate MLP target contrasts.

Target	Difference	95% CI low	95% CI high	BH-adjusted p
AChE	-0.0271	-0.0463	-0.0110	< 0.001
DPP4	0.0045	-0.0015	0.0110	0.261
PTGS2	-0.0005	-0.0063	0.0050	0.900

5 Sensitivity analyses

Table 5: Temporal test macro PR-AUC across five seeds.

Model	Macro PR-AUC
Extra Trees	0.9432 ± 0.0008
Shared MLP	0.9425 ± 0.0024
Logistic regression	0.9257 ± 0.0000
Separate MLP	0.9247 ± 0.0029

Table 6: Continuous pChEMBL test MAE across five seeds; lower is better.

Model	Macro	AChE	DPP4	PTGS2
Separate MLP	0.8083 ± 0.0066	0.7990	0.7306	0.8954
Shared MLP	0.8181 ± 0.0049	0.8153	0.7326	0.9064

6 Natural-product subgroup

The ChEMBL molecule-dictionary natural-product field was joined after applying the same structure standardization as the frozen cohort. The source database identified COCONUT 2025-07 as the flagging resource. Of 14,458 mapped molecules, 657 were flagged; 402 overlapped the primary binary cohort and 79 occurred in its test set.

Eligibility required at least 30 active and 30 inactive observations per target. Because no target passed, `table_natural_product_descriptive.csv` contains descriptive scores only and no hypothesis test.

7 Independent-source eligibility audit

The October 2026 BindingDB article-curated subset was screened before any model score was calculated. Exact human UniProt target identity, a single protein chain, exact numeric Ki or IC50, and a DOI or PMID were required. Documents present in the ChEMBL 37 extraction, exact standardized-molecule overlaps, and candidates with maximum radius-2 Morgan Tanimoto at least 0.70 to any molecule in the ChEMBL modelling cohort were excluded.

Of 1,908 target-matched rows, 1,340 strict binary measurements were available; document filtering left 459 measurements. Removing four contradictory pairs, 93 exact molecule overlaps, and 78 similarity overlaps produced 233 pairs.

Table 7: Natural-product-flagged primary test observations.

Target	Observations	Active	Inactive	Inferentially eligible
AChE	55	11	44	No
DPP4	7	5	2	No
PTGS2	19	9	10	No

Table 8: BindingDB external-candidate pairs after all disjointness filters.

Target	Total	Active	Inactive	Inferentially eligible
AChE	119	32	87	Yes
DPP4	58	35	23	No
PTGS2	56	17	39	No

Eligibility required at least 30 active and 30 inactive pairs for every target. The three-target external cohort therefore failed its locked gate. No external model-performance estimate was calculated. The audit manifest, input hashes, and exact exclusion counts are distributed with the release.

8 Computational environment and quality control

Table 9: Execution environment for the reported runs.

Component	Version
Python	3.11.9
NumPy / pandas / pyarrow	1.26.4 / 2.3.3 / 24.0.0
scikit-learn / SciPy	1.8.0 / 1.17.1
RDKit	2025.09.5
PyTorch / CUDA runtime	2.5.1+cu121 / 12.1
Chemprop	2.2.1
GPU	NVIDIA GeForce RTX 4060 Laptop GPU (8 GB)
Operating system	Windows 10, build 26200

The unit-test suite covered configuration validation, masked losses, nnPU risk, gated attention, split invariants, temporal chronology, validation-only selection, ranking and calibration metrics, and ChEMBL extraction rules. Ruff static checks and `git diff --check` were run before release. Synthetic CUDA smoke outputs are explicitly marked `scientific_results=false` and excluded from all result tables.

9 Machine-readable evidence inventory

The evidence directory contains six vector/raster figure pairs and six CSV tables. Its manifest hashes the complete input run directories and every generated artifact. Table S8 lists compact digest prefixes; the full 64-character SHA-256 values are retained in `docs/evidence/manifest.json`.

Input	SHA-256 prefix

Primary neural	`c571896ebf1f3c92`
Primary classical	`d31d9e61a3af6ed3`
Chemprop D-MPNN	`43d5e0b88b79acb2`
Temporal neural	`e9835378e3954338`
Temporal classical	`a34e97e426fd5987`
Continuous neural	`83e7bb6dcf6a34a7`
Natural-product mapping	`035e54f41bb04383`

10 Locked VietHerb experiment

The implemented plant-bag branch represents each species as a bag of reported metabolites, applies gated attention, and optimizes observed plant uses with non-negative positive–unlabelled risk. The required real-data comparison is activity-only versus naive broadcast versus bag MIL versus shuffled-label and random-label controls under chemical and botanical cold starts. It will not be unlocked until (i) written terms permit a reproducible VietHerb access route and (ii) two domain reviewers complete and adjudicate the therapeutic-use mapping. Accordingly, this supplement contains no estimate for the central ethnopharmacological effect.